

Subject : DS-505: Big Data Handling and Management for Machine Learning Applications
Course : Bachelors of Science in Data Science and Analytics Semester : Semester-5
Model Paper 3

Q.1 A	Answer the following in short (Any 4)	04 Marks
1	Which parameter in <code>pd.read_csv()</code> allows reading a large file in smaller batches to avoid memory crashes? a) <code>batch_size</code> b) <code>chunksize</code> c) <code>step</code> d) <code>nrows_per_block</code>	
2	In Apache Spark, what is the full form of DAG? a) Direct Access Gateway b) Directed Acyclic Graph c) Distributed Action Group d) Dynamic Allocation Grid	
3	In a Decision Tree, a decision node that cannot be split any further and contains the final predicted class is called: a) Root Node b) Internal Node c) Leaf Node d) Branch	
4	Which of the following is an example of an AI task handled by Large Language Models? a) Automated language translation b) Text summarization c) Question answering chatbot d) All of the above	
5	In Scikit-Learn, which method is called to make predictions on test data after training a model? a) <code>.fit()</code> b) <code>.transform()</code> c) <code>.predict()</code> d) <code>.score()</code>	
Q.1 B	Answer the following in short (Any 4)	04 Marks
1	What is meant by the "Velocity" of Big Data? Give one example.	
2	Why does <code>len(df)</code> give an error on a PySpark DataFrame? Which function should be used instead?	
3	What is the mathematical goal of Min-Max Normalization?	
4	What is an ethical consideration or risk when using Large Language Models (AI systems)?	
5	Name the three machine learning models included in the DS-505 syllabus.	
Q.2 A	Write a short note on the following (any one)	05 Marks
1	Explain the characteristics of Big Data (Volume, Velocity, Variety) and discuss why Big Data handling is essential for modern Machine Learning and AI applications.	
2	Explain the Driver Program and Worker Nodes in Apache Spark. What are the main responsibilities of each component in cluster computing?	
B	Explain the difference between Narrow Dependency and Wide Dependency in Apache Spark with simple examples. Why does wide dependency require a network shuffle?	04 Marks
Q.3	Write a short note on the following (any Two)	08 Marks
1	What is the purpose of splitting a dataset using <code>train_test_split()</code> ? Explain what Stratified Splitting is and why it is useful for imbalanced datasets.	
2	Explain the basic steps of Text Cleaning and Preprocessing in Python (such as lowercasing, removing extra spaces, and tokenization) with simple Python code examples.	
3	Explain why Feature Scaling is necessary in Machine Learning. Differentiate between StandardScaler and MinMaxScaler in simple words.	

